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Polymerization of six-membered propylene oxalate

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ABSTRACT

Ring-opening polymerization (ROP) of the six-membered cyclic propylene oxalate was studied for the first time. The reaction proceeded with a high limiting conversion of about 96 % at 100 °C in the presence of tin(II) octoate. Linear macromolecules, Mw of 20–30 × 10^3, Đ ~ 2, represented 86 % of the product, 4 % came from the equilibrium concentration of monomer, while the rest 10 % were cyclic oligomers consisting of 3–10 repeat units. ROP was accompanied by partial (about 30 %) conversion of the catalyst into the insoluble tin(II) oxalate in the course of polymer cross-linking reaction. The kinetics of polymerization obeyed first-order law, indicating typical for ROP constant number of active centers during the process. Polymerization without catalyst was 2000 times slower and also resulted in formation of polymer with Đ ~ 2. Addition of benzyl alcohol to the polymerization mixture resulted in its inclusion into linear macromolecules and a decrease in their molecular weight. The results point to the possibility of synthesizing propylene oxalate block copolymers using macroinitiators with hydroxyl end groups.

1. Introduction

Aliphatic polyesters are widely employed in a variety of fields, including construction materials, packaging, and the apparel industry, as well as implant manufacturing and carriers for drug delivery. Polyoxalates (POx) are oxalic acid-based polyesters. Because of the excellent biodegradability of various polyesters, they are acknowledged as prospective drug carriers [1]. POxs are characterized with a high degradation rate, resulting in the rapid drug release required for different kinds of therapy [2]. On the other hand, a broad functional variety of POx is another advantage in comparison with the most popular drug carriers (PLA, PGA, PLGA, PCL, etc.). Furthermore, the presence of oxalic fragments in the polymer chain causes a specific interaction with metal ions [3] appropriate for radiopharmacology, or reaction with hydrogen peroxide used in cancer therapy [4,5]. POx breakdown produces oxalic acid, which inhibits lactate dehydrogenase and thus suppresses glycolysis in tumor cells [6,7], induces sequestration of calcium in astrocytes that is beneficial in treatment of melanoma metastasis in brain [8] providing further advantages for employing POx carriers in cancer treatment.

Moreover, the need for POx has lately surged since cyclic oxalates, as well as carbonates, can be utilized for the catalytic fixation of atmospheric carbon dioxide, one of the most significant greenhouse gases. The manufacturing of biodegradable plastics from the resultant monomer appears to be a viable solution to the global problem of atmospheric cleanup from greenhouse gasses, which are thought to be responsible for climate change [9,10].

The most widely used methods of polyester production are polycondensation and ring-opening polymerization (ROP) of cyclic monomers containing ester groups. Polycondensation is more widely used in polymer technology to produce structural and functional materials. It represents the main route for the synthesis of POx. W. H. Carothers was the first who proposed synthesis of POx via polycondensation by heating of diols with oxalic esters under reduced pressure [11]. Following studies performed the reaction in two stages: (i) pre-polycondensation by heating diols with oxalic esters under an inert atmosphere and

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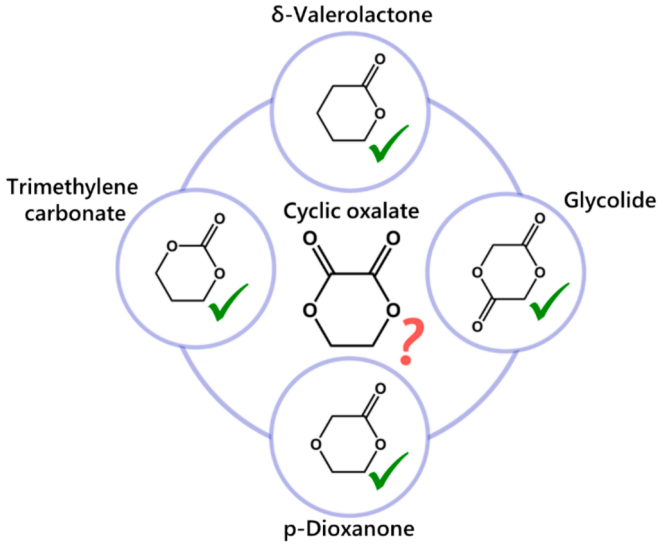
normal pressure, accompanied by alcohol distillation, and (ii) further polycondensation under reduced pressure resulting in formation of a high molecular weight polymer [12–14]. POx can be synthesized by polycondensation of diols with oxalic acid [15,16] or oxalyl chloride [2,17]. However, these reactants can be consumed through decarboxylation or hydrolysis, respectively. Synthesis of POx by polycondensation has inherent disadvantages such as poor control over polymer molecular weight distribution and the difficulty to create polymers of defined architecture, for instance, block copolymers [18].

ROP is another route of POx synthesis. Its efficiency depends on the stability of the cyclic structure of the monomer. More stable monomers are less polymerizable [19]. Thus, 6-membered cycles tend to be less polymerizable than the more strained 3–4-membered rings [20]. Only five types of oxygen-containing 6-membered cyclic esters were polymerized via ROP: δ-valerolactone [21,22], trimethylene carbonate [23,24], dioxanone [25,26], and dimeric glycolide (lactide) [27–30] (Scheme 1). Effect of the nature of catalyst on the structure and polydispersity of the resulting polymers [31,32], the probability of intermolecular and intramolecular transesterification [33], formation of cyclic products [34,35] etc., has been established reliably in experiments with these monomers. The use of homo- [36] and hetero- [37] metallic dinuclear complexes of Al, Zn, Ti, and even Li or Mg facilitate controlling the molecular mass distribution of polyesters produced by the ROP. Cyclic oxalates are an exception from this series of monomers, because the information about their polymerization patterns is limited and fragmentary.

Carothers established the groundwork for ROP of cyclic oxalates [11]. Polyethylene oxalate (PEOx) and polypropylene oxalate (PPOx) were depolymerized to obtain cyclic monomers capable of spontaneous polymerization in air. Research in the field of cyclic oxalate polymerization were almost entirely focused on the synthesis of PEOx. Typically, ethylene oxalate was polymerized by heating at 50–200 °C in the presence of different catalysts [38]. Various inorganic salts and coordination compounds were commonly used for ROP of ethylene oxalate [15,39], for instance, Lewis acids based on tin, antimony, zinc, titanium, and aluminium [38,40,41]. The polymerization of cyclic oxalates is also an intriguing challenge in terms of the activity of the reaction centers. The peculiarity of the oxalate structure is the lowered electron density on carbonyl carbons that makes oxalates highly susceptible to nucleophilic attack by the initiator or the growing chain. This property can influence various steps of ROP.

PPOx hydrolytic breakdown yields low-toxic propylene glycol [42], making this polymer more preferred for drug administration than PEOx,



Scheme 1. Six-membered cyclic esters polymerizable via ROP.

which yields hazardous ethylene glycol. However, ROP of propylene oxalate [11,43,44] and 2-butene oxalate [43] were mentioned only in a few works. No information concerning the polymerization degree, polydispersity, end groups of the polymerization products, the processes by which they are formed, was found in the literature.

In the current research, we looked into the ROP of cyclic propylene oxalate in the presence of tin(II) octoate ($Sn(Oct)_2$). This is the most commonly used catalyst in the ROP of esters [45,46]. We used it as the first-choice catalyst for previously unexplored polymerization of cyclic oxalates. We determined the chemical structure of the main polymerization products and estimated their relative quantities using ^{1}H NMR spectroscopy, size-exclusion chromatography (SEC), and MALDI-TOF (Matrix-Assisted Laser Desorption-Ionization-Time of Flight) mass spectrometry. Our results showed that propylene oxalate in the presence of $Sn(Oct)_2$ underwent polymerization via ROP mechanism forming linear polymers with hydroxyl groups at the chain ends and about 10 % of cyclic oligomers. The polymerization was not affected by decarbonylation, which is typical for the polymerization of cyclic carbonates of 1,2-diols [47]. Cross-linking of polymer chains during the polymerisation caused formation of tin(II) oxalate (SnC_2O_4). The influence of the catalyst content on polymerisation rate and polymer molecular weight distribution was investigated. The possibility of controlling the molecular weight of the polymer by addition of benzyl alcohol as an added initiator and its incorporation into the polymer was demonstrated. These results open up prospects of cyclic oxalates using in the macromolecular design of biodegradable polymers.

2. Experimental

2.1. Materials

Propylene glycol (Sigma-Aldrich, USA) was dried over molecular sieves 4Å and vacuum distilled. Tetrahydrofuran (THF) and diethyl ether (Component-Reaktiv, Russia) were distilled over KOH, refluxed over sodium metal with benzophenone (Sigma-Aldrich), and distilled on the day of the experiment. Ethanol (Component-Reaktiv, Russia) was dehydrated by heating over CaO (ethanol/CaO mass ratio was 3:1) under reflux, distilled, refluxed over molecular sieves 3 Å and redistilled. Benzyl alcohol (Component-Reaktiv, Russia) was distilled under reduced pressure. Isopropanol (Component-Reaktiv, Russia) was dehydrated by heating over Mg powder under reflux and distilled. Oxalyl chloride (Sigma-Aldrich, USA), $Sn(Oct)_2$ (ABCR, Germany), potassium iodide (Fisher Scientific, USA), sodium hydrocarbonate, potassium-sodium tartrate, sodium thiosulfate, iodine crystals (Component-Reaktiv, Russia), and $CDCl_3$ (CarlRoth, Germany) were used as received.

2.2. Synthesis of cyclic propylene oxalate

Cyclic propylene oxalate was synthesized by reaction of oxalyl chloride with propylene glycol according to the procedure initially developed by Iida et al. [48]. We increased ~30-fold the scale of synthesis as compared to the original protocol to facilitate purification of the product. All reagents were dried according to standard procedures and stored under argon. The three-neck flask with drop funnel, thermometer, and magnetic stirring bar was charged with propylene glycol (32.7 g, 0.43 mol), 200 cm^3 of THF, and about 300 cm^3 of pyridine/THF mixture (103.1 cm^3 of pyridine, 1.28 mol, 197 cm^3 of THF). The flask was chilled in the ice-salt bath to 0 °C. Cold solution of oxalyl chloride (43.7 cm^3, 0.51 mol) in THF (300 cm^3) was slowly added through the drop funnel under constant stirring. The solution was added dropwise to prevent the rise of the temperature above 10 °C. The reaction mixture grew murky with the first drops of chloroanhydride. A white, copious precipitate of pyridinium chloride was formed, while the solution turned yellowish. After adding the entire amount of oxalyl chloride, the reaction was allowed to proceed for additional 30 min. Then, THF was removed at 40–45 °C under reduced pressure using rotary evaporator.

The residue was dispersed in 300 cm^3 of chloroform. The mixture was washed twice with equivalent volumes of cold 1 M citric acid solution under constant stirring in the ice bath to avoid rolling and splashing of chloroform caused by neutralization of pyridine. The chloroform layer was dried over anhydrous sodium sulfate. Its concentration under reduced pressure produced orange viscous oil representing the polycondensation product. It was analyzed by SEC for comparison with PPOx obtained by ROP. Subsequent depolymerization the polycondensation product at 130–140 °C under vacuum (around 0.1 Torr) gave a yellowish oil (34.7 g). The yield was 62 % close to 66 % stated in the Ref. [48]. Redistillation of the product under vacuum (0.1 Torr) produced a colorless oil of propylene oxalate (27.4 g, 49 %). It was stored under argon in a desiccator over dry NaOH at 4 °C and was always sampled in the argon stream. ^{1}H NMR ($CDCl_3$, 600 MHz, Fig. S1 in Supplementary Material) $\delta_H = 1.49$ (d, $^3J = 6.6$ Hz, 3H, CH_3), 4.46 (dd, $^2J = 12.7$, $^3J = 9.0$ Hz, 1H, CH_2), 4.53 (dd, $^2J = 12.7$, $^3J = 2.6$ Hz, 1H, CH_2), 4.99 (dqd, $^3J = 9.0, 6.6, 2.6$ Hz, 1H, CH). The spectral data were identical to those published previously [48].

2.3. ROP of cyclic propylene oxalate

Cyclic propylene oxalate was polymerized in the ampoules sealed under vacuum. Prior to polymerization, the ampoules were flame-dried and cooled in a stream of dry argon. Typically, the monomer (100 mg, 0.77 mmol) was placed into the ampoule in a stream of argon connected to the vacuum line and subjected to three pumping out-argon filling cycles at 50 °C. Then 50 μL of $Sn(Oct)_2$ solution in dry THF (0.031 ÷ 0.1 M) was injected into the sample through a rubber septum using gas-tight syringe (Hamilton). In the experiments with the added initiator, an appropriate amount of benzyl alcohol was added to the catalyst solution. The reaction mixture was mixed immediately, the rubber septum was changed for a glass stopper in the stream of argon and the ampoule was frozen in liquid nitrogen. Then THF was removed under vacuum and the ampoule was sealed under reduced pressure. Polymerization was conducted at 100 °C for predefined time interval. During the reaction, the sample solidified and a white precipitate appeared. To isolate the polymer, the glassy mass was dissolved in dry THF (0.5 cm^3). White insoluble powder was removed by centrifugation and washed twice with THF (0.5 cm^3). The supernatant was evaporated to dryness and stored under dry argon in desiccator over NaOH at 4 °C.

2.4. Separation of the polymer from the oligomer fraction

The polymer and oligomer fractions were separated by dissolution of the sample (ca. 60 mg) in dry THF (1 cm^3) and the polymer precipitation with equal volume of diethyl ether. The sediment was isolated by centrifugation (7000×g, 10 min) and dried in the stream of argon.

A similar procedure was used for polymer purification from the catalyst, except a 0.11 M solution of oxalic acid in ether was used instead of pure ether. The precipitate was collected by centrifugation, dissolved in THF and again centrifuged to remove the insoluble SnC_2O_4. The procedure was repeated 3 times. The resulting polymer solution in THF was evaporated in vacuum.

2.5. Synthesis of the reference tin(II) oxalate

SnC_2O_4 was made according to the modified previously published procedure [49]. Briefly, a solution of 2.55 g (0.02 mol) of oxalic acid in 42 cm^3 of absolute ethanol (0.47 M solution) was gently added dropwise through a drop funnel at 50 °C under continuous stirring to the tin(II) chloride solution. Large agglomerates were removed from SnC_2O_4 suspension by filtration through POR100 with pore sizes of 40–100 μm.

2.6. Synthesis of the monoisopropyl oxalate and model reactions with Sn(Oct)$_2$

Monoisopropyl oxalate was obtained according to the previously published procedure [50] by the reaction of isopropanol (3.9 cm^3, 51 mmol) with oxalyl chloride (8.7 cm^3, 101 mmol) in cold diethyl ether (300 cm^3). Crude product (5.5 g, 42 mmol) was distilled under reduced pressure to obtain pure monoisopropyl oxalate (3.9 g, 30 mmol). The substance was characterized by ^{1}H NMR spectroscopy: $\delta_H = 1.39$ (d, 3H, CH_3), 5.20 (sep, 1H, CH_2), 9.95 (s, 1H, OH).

Reaction of monoisopropyl oxalate with $Sn(Oct)_2$ was carried out in the same way as the polymerization. 100 mg of monoisopropyl oxalate (0.76 mmol) and 470 μL of $Sn(Oct)_2$ solution in THF (1.62 M, 0.76 mmol) were mixed in an ampoule, sealed under reduced pressure and kept at 100 °C for 24 h. An abundant white precipitate was formed after removal of the solvent.

PPOx destruction in the presence of $Sn(Oct)_2$ was performed using a similar technique. The unpurified polymer (100 mg) and 100 μL of 0.5 M $Sn(Oct)_2$ in THF were mixed. After removal of THF under vacuum the ampoule was sealed. The reaction was carried out for 24 h at 100 °C.

2.7. General procedures

The molecular weight distribution of polymers was analyzed by SEC at a flow rate of 1 $cm^3 \cdot min^{-1}$ using a system consisting of an Altex pump, Knauer Smartline RI detector, Rheodyne® 7125 Front-Loading Injector, and HR-3 and HR-4 Ultrastyragel columns at 40 °C. The system were calibrated with a set of standard poly(ethylene glycol) samples (Mn = 600, 970, 1500, 1900, 4450, 5000, 6240, 12000, 23100, 44000, and 94000, Waters, USA). Two negative system peaks (1380 and 1500s) were removed from the chromatograms. NMR spectra were acquired on Bruker Avance 600 spectrometer at 303 K; the chemical shifts δ_H were measured in ppm with respect to the solvent ($CDCl_3$, $\delta_H = 7.27$ ppm).

The polymers were analyzed by Matrix-Assisted Laser Desorption-Ionization-Time of Flight (MALDI-TOF) mass spectrometry on UltrafleXtreme MALDI-TOF/TOF mass spectrometer (Bruker Daltonics, Bremen, Germany) to detect positive molecular ions (MH^+ or MNa^+ or MK^+) in the range of 100–6000 *m/z*. For these experiments, 0.5 μL of a sample in THF was mixed with 1 μL of 2,5-dihydroxybenzoic acid solution (40 $mg \cdot cm^{-3}$ in water/acetonitrile 7:3 mixture). The spectra were recorded in reflector mode, and the accuracy of the monoisotopic mass peak measurement was within 0.2 Da. Mass spectra were processed using FlexAnalysis 3.3 software (Bruker Daltonics, Bremen, Germany).

Phase transition temperature in the cyclic propylene oxalate was measured with differential scanning calorimetry using a DSC 823e instrument (Mettler Toledo, Greifensee, Switzerland) in the temperature range −60 – +50 °C (heating rate 10 $K \cdot min^{-1}$) in aluminum 40 μL crucibles.

The precipitate formed during ROP of the cyclic propylene oxalate was identified using elemental analysis according to classical procedures [51]. The results were additionally confirmed by thermal gravimetric analysis with STA 449 F3 Jupiter by Netzsch (Selb, Germany). The measurements were performed in corundum crucibles in the temperature range from 303 to 1073 K at a heating rate of 10 $K \cdot min^{-1}$. The inert gas flow rate (argon) was 70 $cm^3 \cdot min^{-1}$.

The phase composition and crystal structure of tin(II) oxalate were determined by X-ray diffraction (XRD) on the DRON-4–07 diffractometer (Bourevestnik, Russia) with the monochromatic Cu Kα radiation (λ = 1.5406 Å) in 2θ range 20–60° with the step 0.1.

The amount of SnC_2O_4 formed during ROP was determined by iodometric titration of tin(II) in samples after dissolution of the precipitate in HCl according to the protocol [52]. Briefly, SnC_2O_4 was separated from the reaction mixture by centrifugation. White powder was extensively washed with THF and air dried. The pellet was dissolved in 50 μL of concentrated hydrochloric acid and 5 μL of the solution was added to 100 μL of the saturated aqueous solution of $NaHCO_3$ (20 mg)

and potassium-sodium tartrate (10 mg). Then, Sn(II) was directly titrated with a standard 0.1 M I_2-KI solution using starch as an indicator. The method was validated and calibrated with the freshly synthesized SnC_2O_4 (Fig. S2).

Residual tin content in the samples was determined using an inductively coupled plasma atomic emission spectrometer (ICP AES, Agilent 720, Australia). Sample aliquot (about 100 mg) was dissolved in 10 cm^3 of 38 % HCl by boiling under reflux for 1.5 h until complete hydrolysis of the polymer and analyzed according to standard protocol. An aliquot of the reaction mixture was treated in the same way and used as a positive control for Sn determination. 2100 ± 45 ppm of tin was detected in the control sample.

All confidence intervals were calculated assuming a Student's t-distribution. The significance level was set at 0.05.

3. Results and discussion

3.1. Synthesis of cyclic propylene oxalate

Oxalyl chloride and propylene glycol in dry THF underwent polycondensation in the presence of pyridine as a base (Scheme 2a). The product had a very broad molecular mass distribution (MWD) with Mw of about 8000 $g \cdot mol^{-1}$ and Đ = 11 (Fig. 1a, curve 1, Table 1, sample 1). This result is consistent with numerous works, whereby the distribution width of the polycondensation product varied with changes in the mixing conditions of the reactants, the formation of interphase boundaries, the phase state of the reaction mixture, the solvent and even the stirring mode [53,54].

We derived cyclic propylene oxalate from the polycondensation product by pyrolysis according to the procedure originally described by Iida et al. (Scheme 2a) [48]. The monomer was isolated by distillation under high vacuum as a colorless viscous liquid. Its ^{1}H NMR spectrum contained all peaks corresponding to the cyclic propylene oxalate described previously (Fig. S1) [48]. In addition, minor signals shifted upfield relative to the dominant peaks were noticeable at significant amplification. We suggested that they were due to an admixture of propylene glycol monooxalates (at least two compounds that are not equivalent in terms of chemical shifts). They could appear from partial hydrolysis of the monomer. Such transformation led to reduction in the electron density at methine groups and the upfield shift of the proton signals. Vacuum distillation of propylene oxalate decreased the monooxalates amount (Fig. S3a, b). Analysis of thermal transitions in the monomer with differential scanning calorimetry revealed its pronounced crystallization at −35.7 °C with an enthalpy of roughly 3.1 $J \cdot g^{-1}$ (Fig. S4). Low melting point may be due to racemic propylene glycol used herein. The asymmetric carbon atom appears to hinder crystallization of the racemate.

3.2. Ring-opening polymerization of cyclic propylene oxalate catalyzed by $Sn(Oct)_2$

The monomer readily polymerized at 100 °C upon addition of $Sn(Oct)_2$. The reaction proceeded even in the absence of any added initiators. Therefore, we suggested that ROP was initiated by trace amounts of the hydrolyzed monomer (Scheme 2b) that is common for ROP of cyclic esters [55]. Several features were specific for propylene oxalate ROP: (i) formation of noticeable amounts of oligomers, (ii) formation of a precipitate.

3.2.1. SEC and ^{1}H NMR analysis of ROP products

SEC of the reaction mixture after ROP of the cyclic propylene oxalate (Fig. 1a, curve 2) showed two types of products, i.e. oligomers made of 3–10 monomer repeat units along with the unconsumed monomer (t_R = 1280 s), and polymers with Mn ~ 10^4.

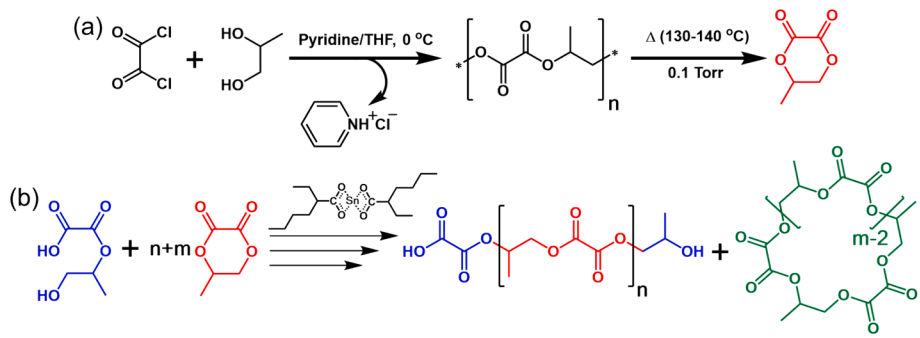
The peak mass of the oligomers was around 500, irrespective of the experimental conditions. On the assumption of equal refractive index increments for all components, the combined mass fraction of oligomers and monomers was about 17 %. Their presence prevented reliable determination of Mn of the linear polymer. Therefore, we calculated the MWD moments by subtracting the oligomer fraction fitted by a Gaussian distribution function from the chromatogram of the reaction mixture (Fig. S5a). Thus, determined polydispersity of the linear polymer appeared to be about 2 (Table 1, sample 2). According to Flory theory, such polydispersity index, Đ, at equilibrium state could result from intermolecular transesterification [56]. It is a common side process in the synthesis of polyesters, for instance, L-lactide polymerization at equilibrium [57]. Similarity of the Đ value determined herein with the predicted theoretically indicated the achievement of the equilibrium in transesterification. PPOx obtained by ROP contained substantially less oligomers (Fig. 1a, curve 2) than that obtained by polycondensation (Fig. 1a, curve 1).

Thus, the advantages of using ROP became obvious, since it allowed obtaining polymers with Mn = 10900 $g \cdot mol^{-1}$ and Đ = 2.01 (Fig. 1a).

The polymer purification by precipitation with diethyl ether from a THF solution dramatically reduced the monomer and oligomer content (Fig. 1b). The SEC trace of the purified polymer nearly matched the curve after subtraction (Fig. S5b), indicating the accuracy of the oligomer signal subtraction.

Comparison of the ^{1}H NMR spectra of the reaction mixture (Fig. 2) and monomer (Fig. S1) showed considerable shifts of main signals caused by ring opening. As a result of ring strain relief, the signals of methyl and methylene protons shifted upfield, while the signals of methine protons shifted downfield. Opposite shifts of proton signals were described for ROP of different cyclic esters [58,59].

The peak of the methine protons of the polymer had a shoulder centered around 5.40 ppm and most likely corresponding to the methine protons of oligomers. Fractional precipitation of the polymer with



Scheme 2. (a) Synthesis of cyclic propylene oxalate monomer and (b) its ROP in the presence of $Sn(Oct)_2$. One of propylene glycol monooxalates (initiators) is shown in blue, cyclic propylene oxalate in red, cyclic oligomers – green. The linear PPOx was the main ROP product. n and m indicate the numbers of repeat units in the linear polymer and the cyclic oligomers, respectively.

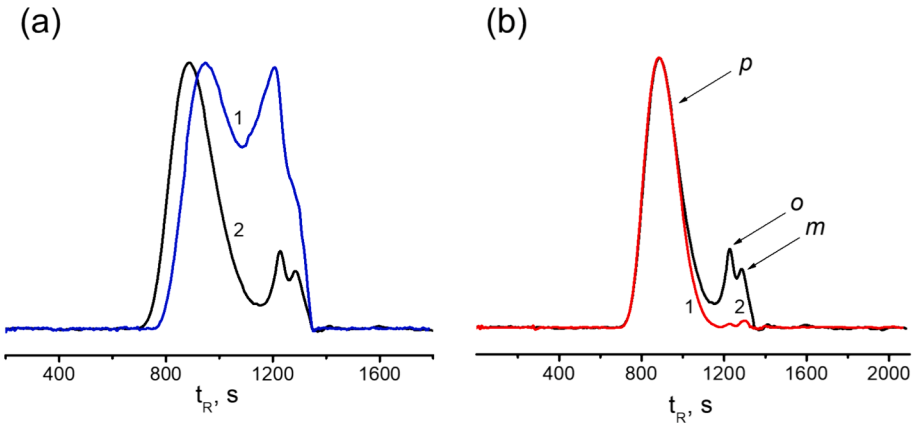


Fig. 1. SEC traces of (a) the product of oxalyl chloride and propylene oxalate polycondensation (curve 1), and the product of propylene oxalate ROP (100 °C, 24 h, C(Sn(Oct)$_2$)$_0$/C(M)$_0$ = 0.0065), (curve 2); (b) the purified (curve 1) and initial (curve 2) samples. The indices *p*, *o*, and *m* denote polymer, oligomer, and monomer peaks, respectively.

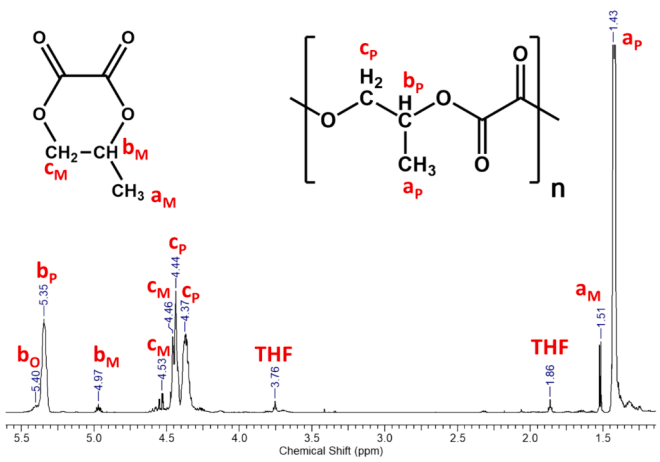


Fig. 2. ^{1}H NMR spectrum of the reaction mixture with referenced signals. The indices b_M, c_M, a_M denote the signals of methine, methylene, and methyl signals of the monomer, b_P, c_P, and a_P, denote the corresponding signals of the same groups in linear macromolecules, respectively. b_O denotes methine signals of the oligomers.

diethyl ether 10-fold decreased the intensity of signal at $\delta_H = 5.40$ ppm (Fig. S6a), while the supernatant became highly enriched with this side product (*cf.* black curves in panels (a) and (b) in Fig. S6). Hence, the peak at $\delta_H = 5.40$ ppm corresponded to the oligomers that were not or only marginally precipitated by equivalent volume of ether. Their mass fraction estimated by deconvolution of ^{1}H NMR signals in the region of δ_H 5.2–5.5 ppm (Fig. S7a) was found to be 10 ± 4 %.

^{1}H NMR data were used for estimation of the polymerization conversion, α, from the intensities of well-resolved with respect to each other peaks of methine protons at $\delta_H = 4.97$, 5.35, and 5.40 ppm (Fig. 2, b_M, b_P, b_O) according to equation:

$$\alpha = \frac{I(5.35) + I(5.40)}{I(4.97) + I(5.35) + I(5.40)} \times 100\% \quad (1)$$

Equilibrium conversion of bulk polymerization of propylene oxalate at 100 °C was about 96 %. Hence, 3.8 ± 0.4 % of the monomer remained unconsumed (Fig. S7b). The same conversion was observed for ROP carried out for one or three days testifying the achievement of equilibrium state. The total amount of the oligomers and the monomer in the reaction mixture was about 14 ± 4 % as judged from ^{1}H NMR data. The result was close to 17 % obtained by the SEC within the experimental error.

The peculiarities of the oligomers and polymer structures were disclosed with MALDI-TOF mass spectrometry.

3.2.2. *MALDI-TOF analysis of the oligomer and polymer fractions*

The spectra of the reaction mixtures after polymerization showed two major groups of signals (Fig. 3, C and L) whose masses grew in steps of 130 (the molecular mass of the propylene oxalate). The absence of a series of peaks with a periodicity different from 130 indicated that the polymerization process did not involve decomposition processes resulting in the formation of ether or polycarbonate fragments. The stability of cyclic propylene oxalate during polymerization differed from the previously reported decomposition of a considerable fraction of 1,2-diol carbonates during ROP [47].

One of the series of peaks consisted of pronounced signals relevant to masses divisible by 130 with the residue of 23 (Na^+) or 39 (K^+). The absence of any other residue indicated that no initiator was included in

Table 1
Characteristics of polymer samples.

Sample	$\frac{C(BnOH)}{C(M)}$	$\frac{C(Sn(Oct)_2)}{C(M)}$	t, min	α[a]	Mn[b]	Mw[b]	Đ[b]	Mass fraction of oligomers, %
1	Polycondensation				690	7600	11.02	9.7
2	0	0.0065	1440	0.96	10900	21900	2.01	7.5
3	0.006	0.0065	1440	0.96	7100	13700	1.93	6.7
4	0	0.0065	40	0.95	9900	19700	1.99	9.2
5	0	0.005	32	0.93	9700	19300	1.99	7.3
6	0	0.0035	35	0.93	9000	19000	2.12	10.1
7	0	0.002	70	0.91	8100	18000	2.21	6.9
8	0	0	36000	0.85	6800	15200	2.22	3.7

[a] α is the degree of monomer conversion determined by NMR as illustrated in Fig. S7b using equation (1) below.
[b] Calculated by subtracting the oligomer fraction fitted by a Gaussian distribution function from the SEC traces of the reaction mixture as shown in Fig. S5.

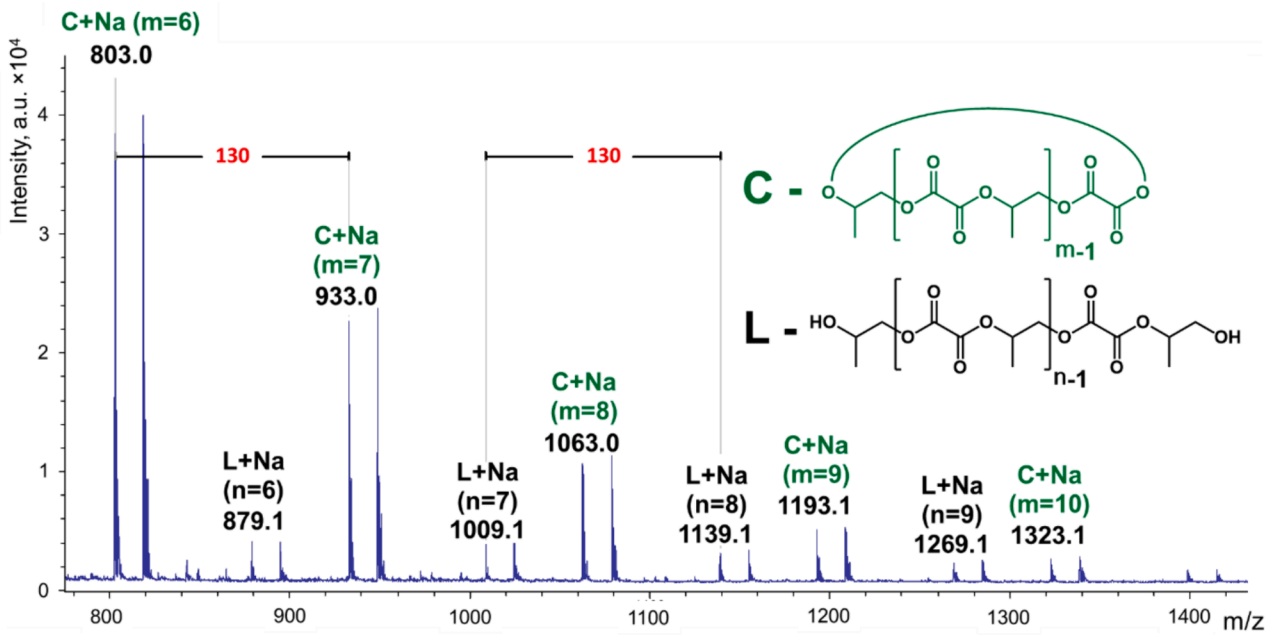


Fig. 3. MALDI-TOF spectrum of the reaction mixture after polymerization of propylene oxalate (100 °C, 24 h, $C(Sn(Oct)_2)_0/C(M)_0 = 0.0065$) in the absence of any additional initiator. The assumed structures of compounds corresponding to the basic modes are marked as "**C**" for cyclic oligomers and "**L**" for linear macromolecules.

this series of compounds. Hence, they were cyclic molecules without end groups (Fig. 3, signals C). Most likely, these compounds dominating in the low molecular weight region of mass-spectra (Fig. S8) corresponded to the oligomeric fraction revealed by SEC analysis (Fig. 1b). The peak intensity of these oligomers exponentially decreased with the increase in the number of the repeat units (Fig. 3, signals C). This tendency was in full accordance with Jacobson and Stockmayer theory, which predicts a power-law decrease in the number of cyclic products as their degree of polymerization, m, increases during the synthesis of polyester via polycondensation. According to the theory, at the equilibrium of transesterification processes, the exponent is −5/2. [60] The degree exponent in our investigations showed sample-dependent variation between −1.8 and −3.8, further validating the cyclic character of this fraction. Intramolecular transesterification (Scheme S1) is the process that forms cyclic oligomers and has been reported in several situations, including polycondensation of dicarboxylic acids with diols [61], ROP of cyclic carbonates [62], p-dioxanone [63], and to a lesser extent lactide [34,64,65]. In particular, polymerization of lactide catalyzed by $Sn(Oct)_2$ produced only trace amounts of cyclic oligomers [57]. Presumably, significant formation of cyclic oligomers during propylene oxalate polymerization was due to higher PPOx activity in transesterification compared to PLA and other polyesters of hydroxycarboxylic acids. Such enhanced activity could be caused by specific electron structure of the oxalic group.

The second group of signals (Fig. 3, signals L) corresponded to linear macromolecules. Their molecular masses were devisable by 130 with the residue of 23 (or 39) + 76 (M=130·n + 76 + 23(39)). The residue 76 equals to the molecular mass of propylene glycol evidently emerging from the propylene glycol monooxalate served as initiator (Scheme 2b). The signals corresponding to linear macromolecules with end groups other than propylene glycol were not found in the MALDI-TOF spectra of the product.

According to ROP mechanism, the amount of propylene glycol and oxalic acid end groups should be equal in the absence of side reactions. However, we did not observe terminal oxalic group in the linear macromolecules. Searching for possible reasons for this fact, we analyzed the contribution of the formation of inorganic precipitate in the end-group imbalance.

3.2.3. *Nature of the precipitate and chemistry underlying its formation*

The white crystalline precipitate formed during ROP was insoluble in organic solvents but soluble in concentrated HCl. Elemental analysis of the precipitate revealed H, C, and Sn in proportions 1.4 ± 0.3, 11 ± 3, 59 ± 3 % which were close to that of SnC_2O_4 (theory: 0, 11.6, 57.4 %). To testify the structure of this precipitate, we compared its X-ray diffraction (XRD) and thermogravimetric (TGA) patterns to those of the reference tin oxalate made by inorganic synthesis.

The XRD pattern of the "polymerization" precipitate coincided with that of the synthesized SnC_2O_4 although had a smoother appearance (Fig. 4a). Presumably, the smoothness of the diffractogram is caused by smaller size of the crystalline domains formed in the extremely viscous environment of the polymer bulk. Both samples contained a single SnC_2O_4 phase (JCPDS 14-0743).

The mass loss curves (TGA) of the synthesized SnC_2O_4 (Fig. 4b, black curve) and the material isolated from the polymerization mixture (Fig. 4b, red curve) were similar. Both reached the plateau at $T \geq 350$ °C that matched the interval 320–370 °C reported previously [66,67]. Some differences between the thermograms of the "polymerization" oxalate and the reference sample could result from the presence of polymer or solvent impurities in the inorganic precipitate and extremely tiny size of the crystals.

The similarity of XRD and TGA of both samples testified that the precipitate formed during ROP was SnC_2O_4. Its amount in the final product of ROP was determined by iodometric titration of Sn^{2+} ions as specified above (section 2.7). To standardize the titration protocol, freshly prepared anhydrous SnC_2O_4 sample was used as reference. The calibration line (Fig. S2) demonstrated a definite proportionality between the amount of SnC_2O_4 in the sample and the amount of KI_3 consumed for its titration. The results showed that up to 30 % (depending on the monomer purity) of tin precipitated as oxalate during polymerization within 24 h. Thus, a part of the catalyst $Sn(Oct)_2$ converted to SnC_2O_4, which formed white powder in the polymer/monomer mixture.

Formation of SnC_2O_4 could occur by $Sn(Oct)_2$ interaction with terminal oxalate groups (Scheme 3a, b), internal groups of the polymers or oligomers (Scheme 3c, d), and with the monomer (not shown).

All reactions shown in Scheme 3 led to alterations in end group distribution. Transesterification reactions involving 2-ethylhexanoic

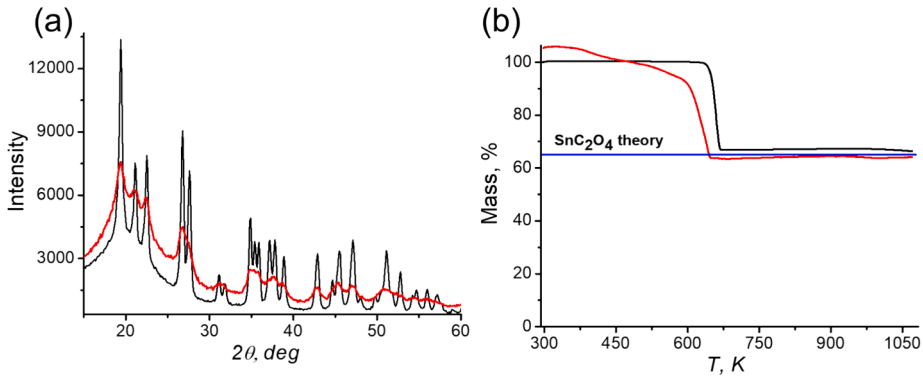


Fig. 4. (a) X-ray diffractograms and (b) TGA curves of the synthesized SnC_2O_4 (black curve) and the precipitate formed during polymerization (red curve). The theoretical level of mass loss during decomposition of SnC_2O_4 under an inert atmosphere is shown in blue.

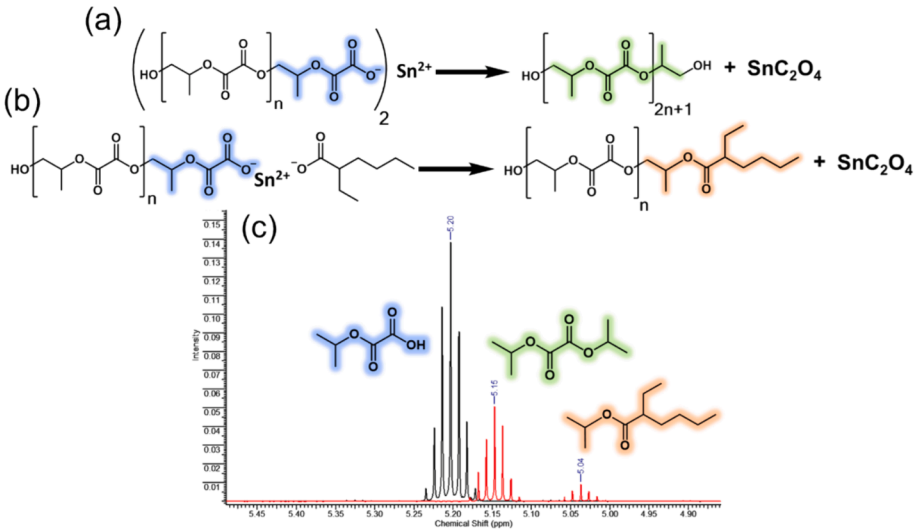


Fig. 5. (a) Probable reactions of SnC_2O_4 formation during ROP or model experiment (highlighted) which involve two oxalic terminal groups; (b) ethylhexanoic acid and oxalic terminal group; (c) the fragments of ^{1}H NMR spectra corresponding to methine protons of isopropyl radicals of monoisopropyl oxalate (black) and products of its reaction with $Sn(Oct)_2$ (red) at equimolar ratio at 100 °C, 24 h. The corresponding structures of products are shown above the signals.

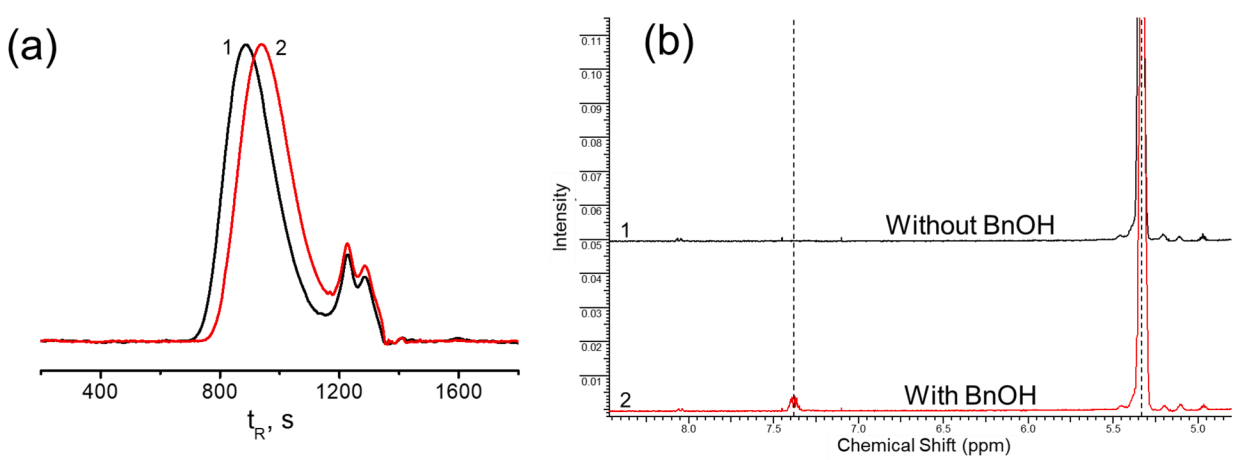


Fig. 6. Influence of benzyl alcohol on the cyclic propylene oxalate polymerization. (a) SEC traces and (b) ^{1}H NMR spectra of PPOx synthesized in the absence (curves 1) and in the presence (curves 2) of 0.6 % of benzyl alcohol at 0.65 % of $Sn(Oct)_2$ in relation to the monomer. Dashed lines indicate the chemical shifts of benzyl (7.38 ppm) and methine (5.35 ppm) groups.

acid (Scheme 3b, c) seemed to be unlikely since no terminal octoates were detected in linear polymers by MALDI-TOF spectrometry (Fig. 3). The reaction with oxalic groups located inside the chain (Scheme 3d) could lead to scission of the macromolecule into two chains with unsaturated enol esters end groups, which are unstable and can undergo further transformations. In contrast, cross-linking of two chains with

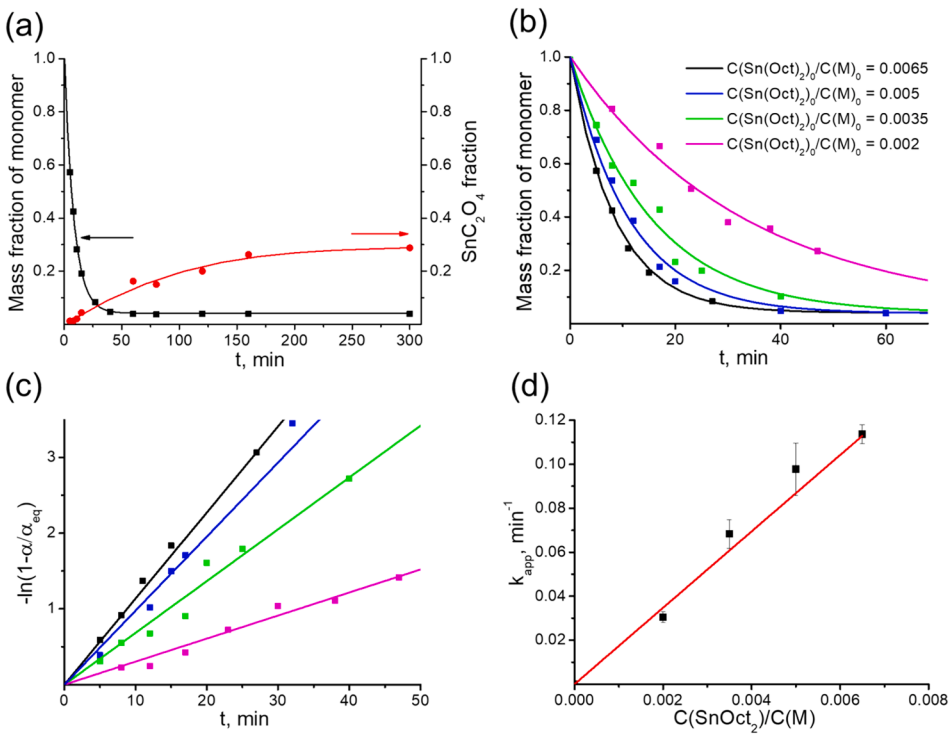
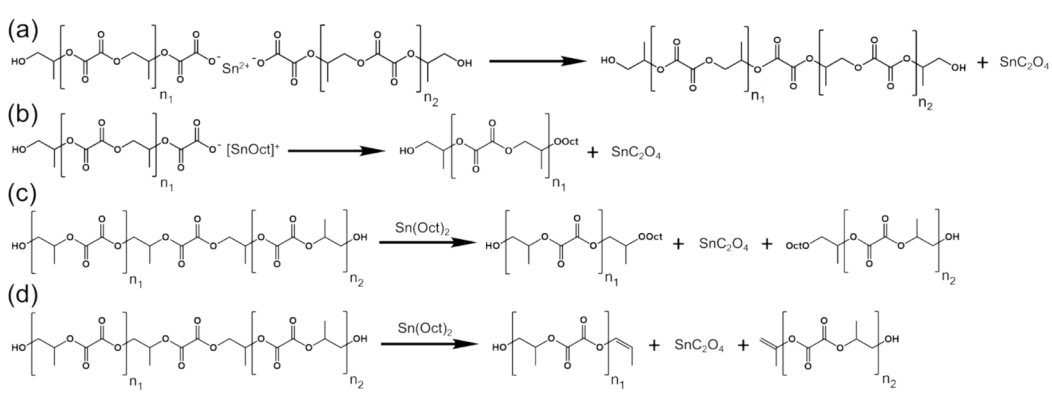


Fig. 7. (a) Kinetics of propylene oxalate consumption (black) and SnC_2O_4 formation (red) during the polymerization catalyzed by $Sn(Oct)_2$ (T=100 °C, $C(Sn(Oct)_2)_0/C(M)_0$ = 0.0065); (b) kinetic curves at different $Sn(Oct)_2$ concentrations: $C(Sn(Oct)_2)_0/C(M)_0$ – 0.0065 (black), 0.005 (blue), 0.0035 (green), 0.002 (magenta) in linear and (c) semi-logarithmic coordinates, α and α_{eq} – current and equilibrium degrees of monomer conversion; (d) dependence of the apparent rate constant on $Sn(Oct)_2$ concentration.



Scheme 3. (a, b) Possible reactions of $Sn(Oct)_2$ with end groups; (c, d) the internal fragments of the polymer leading to SnC_2O_4 formation.

detachment of oxalate end groups shown in Scheme 3a led to formation of a long macromolecule with propylene glycol residues at both ends. Indeed, such polymers were the most abundant ROP product.

Thus, formation of SnC_2O_4 precipitate could be a reason for the reduction of the fraction of terminal oxalates.

We checked the possibility of cross-linking of two chains catalyzed by $Sn(Oct)_2$ in a model experiment. Monoisopropyl oxalate as a model of the terminal group of PPOx was mixed with an equivalent amount of $Sn(Oct)_2$ (Fig. 5a, b). Examination of the reaction products with ^1H NMR (Fig. 5c) showed formation of diisopropyl oxalate (major product) and isopropyl 2-ethylhexanoate (minor product). The result indicates that the cross-linking reaction between terminal oxalic groups could occur in the polymerization mixture (Scheme 3a). It is consistent with the analysis of PPOx with MALDI-TOF spectrometry (Fig. 4). Formation of minute amounts of isopropyl-2-ethylhexanoate indicates that the reaction in Scheme 3b was also possible, but less pronounced.

The possibility of PPOx scission in the presence of $Sn(Oct)_2$ (Scheme 3c, d) was demonstrated by the shift of the polymer chromatogram towards lower molecular weights after incubation of PPOx with $Sn(Oct)_2$ at 100 °C (Fig. S9). Both experiments were accompanied by abundant formation of SnC_2O_4. Thus, these experiments present experimental evidence for the routs shown in Scheme 3.

Two propylene glycol end fragments could also result from decarboxylation of oxalic monoesters and subsequent decarbonylation of formates. Both reactions were reported for ROP of polycarbonates [23]. These reactions are catalyzed by heavy metals and proceed at a relatively high rate at about 100 °C [68,69].

3.2.4. *ROP of propylene oxalate initiated by benzyl alcohol*

Alcohols are commonly used initiators of the polymerization of cyclic esters. Herein benzyl alcohol was used for initiation of propylene oxalate ROP. The addition of 0.6 % (mol.) of benzyl alcohol to the

monomer resulted in a decrease in the molecular weight of the polymer from 10900 g/mol to 7100 $g \cdot mol^{-1}$ (Fig. 6a, Table 1, samples 1, 3). The effect was associated with an increase in the number of polymer chains due to the initiation of polymerization by benzyl alcohol.

Analysis of the ^1H NMR spectrum of the sample obtained in the presence of benzyl alcohol and purified from free alcohol by ether precipitation showed the presence of benzyl radicals in the polymer (Fig. 6b, signals at 7.32–7.43 ppm). Thus, benzyl alcohol was capable to initiate polymerization of propylene oxalate in presence of $Sn(Oct)_2$ that provides the control of the molecular mass of PPOx along with fruitful applications related to synthesis of block-copolymers or functionalization of the resulting polymers.

3.2.5. Polymerization kinetics

Transformation of catalyst to SnC_2O_4 should greatly affect the polymerization kinetics. However, the monomer consumption obeyed first-order kinetics up to limiting conversion

(Fig. 7a-c, black curves, α, α_{eq} – current and equilibrium degrees of conversion). Investigation of SnC_2O_4 formation showed that this process was much slower than monomer consumption and only about 10 % of the catalyst was converted to SnC_2O_4 when the degree of polymerisation conversion reached 90 % (Fig. 7a). Therefore, these processes were spaced out over time and the monomer was unlikely to be involved in SnC_2O_4 formation. Despite the reduced SnC_2O_4 formation, the polymer formed in 40 min exhibited a broad molecular mass distribution with Đ = 1.99 and Mn = 9900 (Table 1, sample 4). This meant that SnC_2O_4 formation was not the main reason for MWD broadening.

Fig. 7a showed that the concentration of the retained monomer approached equilibrium level within 40 min. However, 23 h later the number-average molecular weight of the product was found to be increases from 9900 to 10900 (Table 1). This enlargement of the polymer could be a consequence of the chain cross-linking mediated by $Sn(Oct)_2$, accompanied by SnC_2O_4 formation. Indeed, significant accumulation of SnC_2O_4 went on after the achievement of the polymerization equilibrium (Fig. 7a, red curve). Thus, these results support our assumption about SnC_2O_4 formation due to chain cross-linking.

In contrast to SnC_2O_4, the oligomers formed from the very beginning of the polymerization. The sample isolated at only 20 % degree of conversion contained considerable amount of the oligomers as judged from NMR spectrum (Fig. S10a). The total amount of oligomers increased during the polymerisation, reaching a plateau at a concentration of approximately 10 % (mas.) (Fig. S10b). This plateau indicated equal rates of formation of oligomers due to intramolecular transesterification and their consumption caused by their attachment to the growing polymer chains. This equilibrium was achieved at a monomer conversion degree of 40–50 %, indicating a rapid intramolecular transesterification.

The concentration of the catalyst in the reaction mixture determined the rate of monomer consumption (Fig. 7b). Linearization of the curves in semi-logarithmic coordinates (Fig. 7c) indicated the first-order law of propylene oxalate polymerization. Hence, the number of active centres did not noticeably change during the process even at the reduced catalyst concentration. The permanence of active species concentration is typical for ROP mechanism. The apparent rate constant increased linearly with the catalyst content within the concentration range from 0.2 to 0.65 % (mol.) (Fig. 7d). The turnover frequency (TOF) of $Sn(Oct)_2$ in propylene oxalate ROP was 1000 ± 250 $mol(M) \bullet mol(Sn)^{-1} \bullet h^{-1}$, indicating moderate efficiency of catalysis. Nevertheless, ROP of propylene oxalate proceeded at low concentrations of $Sn(Oct)_2$.

These results seem to be important for the potential perspective of biomedical applications of PPOx [1,2]. However, tin is toxic to humans. At levels above 250 mg per kg of food consumed, tin causes a decrease in the iron content in erythrocytes [70]. Keeping this in mind, we determined the amount of tin in the polymer samples using ICP AES. We found that the polymers purified by three-fold re-precipitation (see 2.4 section) contained 670 ± 49 mg of tin per kg of the polymer. Taking into account that drug carriers are usually administered as aqueous 1–5 % dispersion, the tin content in such formulation would be ≤ 33 ppm, which is practically safe.

The average polymers dispersity for high degrees of conversions (α > 90 %) was about 2 irrespective of the catalyst content (Table 1, samples 2, 4–7). The content of oligomers did not depend on the concentration of a catalyst. Mn of the polymer gradually decreased with a reduction of catalyst content thus confirming SnC_2O_4 induced end groups cross-linking.

Estimation of the Mn of polymers obtained at the moderate degrees of conversion by subtracting of oligomer signal was impossible due to strong overlap between the polymer and oligomer modes (Fig. S11a). Therefore, we were unable to get a dependence of Mn of the linear polymer on the monomer conversion. At the same time, fractionation of the polymerization mixture containing truncated PPOx resulted in a significant shift of the polymer MWD towards higher molecular weights (Fig. S11b). The most probable polymer mass Mp can be employed to follow the polymer growth during the polymerization as the tangible experimental parameter. It increased gradually with conversion (Fig. S12). The deviation from the theoretically expected linear dependence may be due to either a change in MWD or oligomer content with conversion. Nevertheless, the data obtained indicate that the polymerization of propylene oxalate obeys the basic ROP patterns known for other cyclic esters.

Broadening of MWD can be either an inherent property of propylene oxalate polymerization or the result of catalyst action. To distinguish between these two possibilities, we have studied the polymerization of propylene oxalate in the absence of a catalyst at 100 °C. In this case, the reaction achieved conversion of about 85 % within 25 days (Fig. S13), which is about 2000 times longer than polymerization in the presence of $Sn(Oct)_2$ ($C(Sn(Oct)_2)_0/C(M)_0 = 0.0065$). Thus obtained polymer was characterized by comparatively low Mn = 6800 $g \cdot mol^{-1}$ at Đ ~ 2 and contained about 4 % of oligomers. (Table 1, sample 8). As expected, the ROP without catalyst led to the product with Mn considerably less than those obtained in the presence of $Sn(Oct)_2$, which mediates end groups cross-linking (Scheme 3a).

The polymers generated through spontaneous ROP and in the presence of $Sn(Oct)_2$ had similar MWDs. This fact indicates that the catalyst had only marginal influence on polymerization control. Thus, SnC_2O_4 causes an increase in the molecular weight of PPOx, most likely through a cross-linking reaction, but has no effect on MWD.

4. Concluding Remarks

In this study, we investigated for the first time the ring-opening polymerization of propylene oxalate – the only unexplored 6-membered oxygen-containing ROP polymerizable monomer. Unexpectedly, this monomer polymerized readily and with high equilibrium conversion 96 % (mas.) when tin(II) octoate was used as a catalyst. At the same time, unlike ROP of propylene carbonate, the polymerization of propylene oxalate does not involve decarboxylation reactions, and no polypropylene oxide fragments are produced.

ROP was initiated by the hydrolyzed monomer present in the cyclic monomer sample (Scheme 2), or by added initiators comprising OH-containing species, e.g. benzyl alcohol (Scheme S2a). Further propagation in the absence of added initiator led to linear macromolecules with the number-average polymerization degree of 90 – 100 repeat units (Mn ~ 11000 $g \cdot mol^{-1}$) and exhibiting a broad molecular weight distribution with Đ ~ 2 (Scheme S2b).

Some of these molecules may enter an intramolecular transesterification leading to the formation of cyclic oligomers (Scheme S2c) composed of 3–10 repeat units. Their amount decreased rapidly with increasing cycle size (Fig. 3).

The mass fraction of oligomers increased up to 10 % and equilibration with achievement of plateau occurred at monomer conversion of about 40 – 50 % (Fig. S10b). We assume that the oligomers formed

during further polymerization are consumed in the synthesis of linear macromolecules (Scheme S2c). Similar processes have already been reported in the ROP of other six-member cyclic esters [63,64,69,71].

The width of the molecular weight distribution of linear polymers (Đ ~ 2, Table 1) corresponds to the equilibrium of transesterification reactions [56]. Furthermore, in addition to the intramolecular transesterification shown in Scheme S2b, intermolecular transesterification also occurred (Scheme S2d). Spontaneous polymerization in the absence of a catalyst led to similar values of Đ and was also accompanied by the formation of macrocycles. We can therefore conclude that transesterification is an inherent feature of the polymerization of 6-membered cyclic oxalates.

Monomer consumption in the presence of $Sn(Oct)_2$ obeyed the first-order kinetics indicating the preservation of the amount of active centers throughout the process. ROP of propylene oxalate in the absence of catalyst allowed to ascertain that the rate of polymerization was 2000-fold lower than in the presence of 0.65 % (mol.) of $Sn(Oct)_2$, showing a relatively high catalysis efficiency.

Another feature of propylene oxalate polymerization in the presence of tin(II) octoate is the formation of tin oxalate (SnC_2O_4). This occurs due to cross-linking reaction between two chains terminated with oxalate groups (Scheme S2e) that was supported by the increase in PPOx molecular mass along with catalyst concentration. A model experiment on the interaction of monoisopropyl oxalate with tin(II) octoate to form diisopropyl oxalate clearly indicates that such a reaction is possible. At the same time, as shown in Fig. 7a, this process occurred in parallel with polymerization, being considerably slower than chain propagation. Therefore, chain cross-linking does not contribute significantly to the MWD width.

Addition of benzyl alcohol into the polymerization mixture as an initiator led to its incorporation into the polymer resulting in a decrease in the molecular weight of the polymerization product. This result opens up the possibility of obtaining polyoxalates with functional groups at the end of the chain or block copolymers.

CRediT authorship contribution statement

Nikolai P. Iakimov: Writing – original draft, Investigation, Formal analysis, Data curation, Conceptualization. **Ekaterina M. Budynina:** Writing – review & editing, Formal analysis, Data curation. **Anna K. Berkovich:** Writing – review & editing, Methodology, Data curation. **Marina V. Serebryakova:** Writing – review & editing, Methodology, Data curation. **Vadim B. Platonov:** Methodology, Data curation. **Evgenii O. Fomin:** Writing – review & editing, Investigation. **Anastasia G. Buyanovskaya:** Methodology, Data curation. **Ivan V. Mikheev:** Methodology, Data curation. **Nikolay S. Melik-Nubarov:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.eurpolymj.2024.113410.

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